W4D4 **18-1-2026**
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PROGETTO FINALE

Esercizio :

Simulare, in ambiente di laboratorio virtuale, un architettura client server in cui un client con indirizzo 192.168.50.102 richiede tramite web browser una risorsa all'hostname epicode.internal che risponde all'indirizzo 192.168.50.100 (Kali).

Si intercetti la comunicazione con wireshark, evidenziando i MAC address di sorgente e desitnazione ed il contenuto della richiesta HTTPS.

Ripetere l esercizio modificando l'esercizio con la richiesta HTTP, e evidenziare le principali differenze.

Sviluppo

1) Bisogna verificare che le macchine virtuali siano correttamente configurate con i loro IP statici e siano in grado di comunicare tra loro.

```
C:\Users\user>ping 192.168.50.100

Esecuzione di Ping 192.168.50.100 con 32 byte di dati:
Risposta da 192.168.50.100: byte=32 durata=1ms TTL=64
Risposta da 192.168.50.100: byte=32 durata=1ms TTL=64
Risposta da 192.168.50.100: byte=32 durata=2ms TTL=64
Risposta da 192.168.50.100: byte=32 durata<1ms TTL=64

Statistiche Ping per 192.168.50.100:
    Pacchetti: Trasmessi = 4, Ricevuti = 4,
    Persi = 0 (0% persi),
Tempo approssimativo percorsi andata/ritorno in millisecondi:
    Minimo = 0ms, Massimo =  2ms, Medio =  1ms
```

Come possiamo vedere, le nostre macchine (che erano state precedentemente configurate per altri esercizi con IP statici), sono attualmente in grado di comunicare tra di loro.

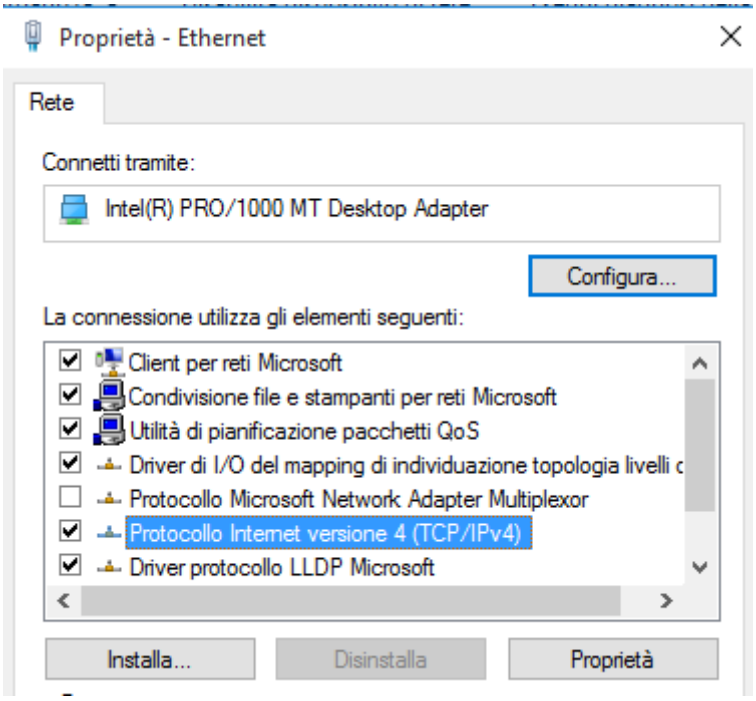
```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ sudo inetsim
[sudo] password for kali:
INetSim 1.3.2 (2020-05-19) by Matthias Eckert & Thomas Hungenberg
Using log directory:      /var/log/inetsim/
Using data directory:     /var/lib/inetsim/
Using report directory:   /var/log/inetsim/report/
Using configuration file: /etc/inetsim/inetsim.conf
Parsing configuration file.
Warning: Unknown option 'EPICODE' in configuration file '/etc/inetsim/inetsim.conf' line 246
Warning: Unknown option 'epicode.internal' in configuration file '/etc/inetsim/inetsim.conf' line 265
Warning: Unknown option 'epicode.internal' in configuration file '/etc/inetsim/inetsim.conf' line 372
Configuration file parsed successfully.
=== INetSim main process started (PID 53670) ===
Session ID:     53670
Listening on:   192.168.50.100
Real Date/Time: 2026-01-18 07:17:06
Fake Date/Time: 2026-01-18 07:17:06 (Delta: 0 seconds)
 Forking services...
  * dns_53_tcp_udp - started (PID 53672)
Can't locate object method "main_loop" via package "Net::DNS::Nameserver" at /usr/share/perl5/INetSim/DNS.pm line 69.
  * https_443_tcp - started (PID 53674)
  * http_80_tcp - started (PID 53673)
 done.
Simulation running.
```

2) A questo punto dopo aver abilitato la nostra rete internet virtuale sulla macchina kali, attraverso il comando “ **sudo inetsim** “, ci assicuriamo che i servizi che a noi interessano per l esercizio siano attivati, per la precisione: DNS , HTTP ; HTTPS

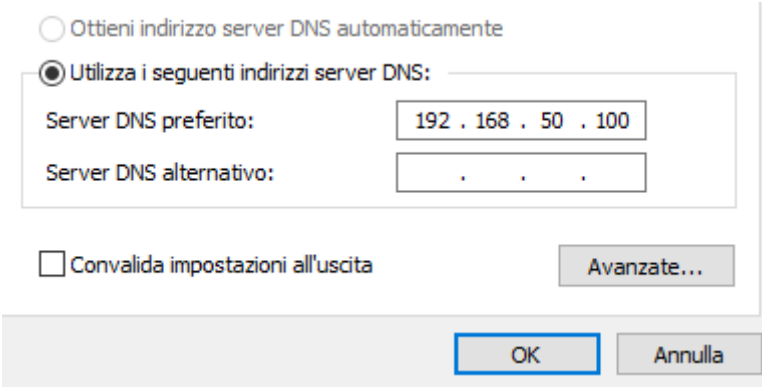
```
#
# Available service names are:
# dns, http, smtp, pop3, tftp, ftp, ntp, time_tcp,
# time_udp, daytime_tcp, daytime_udp, echo_tcp,
# echo_udp, discard_tcp, discard_udp, quotd_tcp,
# quotd_udp, chargen_tcp, chargen_udp, finger,
# ident, syslog, dummy_tcp, dummy_udp, smtps, pop3s,
# ftps, irc, https
#
start_service dns
start_service http
start_service https
#start_service smtp
#start_service smtps
#start_service pop3
#start_service pop3s
#start_service ftp
#start_service ftps
#start_service tftp
#start_service irc
```

il comando per accedere alla Inetsim configuration file è:
“ **sudo nano /etc/inetsim/inetsim.conf** “

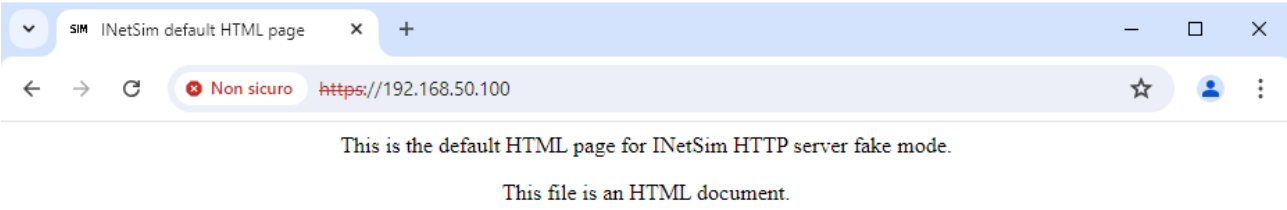
3) Configurazione delle preferenze di rete di Windows, ovvero impostiamo come DNS preferito l indirzzo IP di Kali, in questo modo quando faremo delle richieste HTTP e HTTPS, Windows si metterà in comunicazione sulla relativa porta del nostro server virtuale che abbiamo abilitato su Kali con il servizio di Inetsim.



Nelle impostazioni della scheda di rete su Windows, apriamo le impostazioni del protocollo internet e inseriamo, l indirizzo di Kali come DNS preferito.



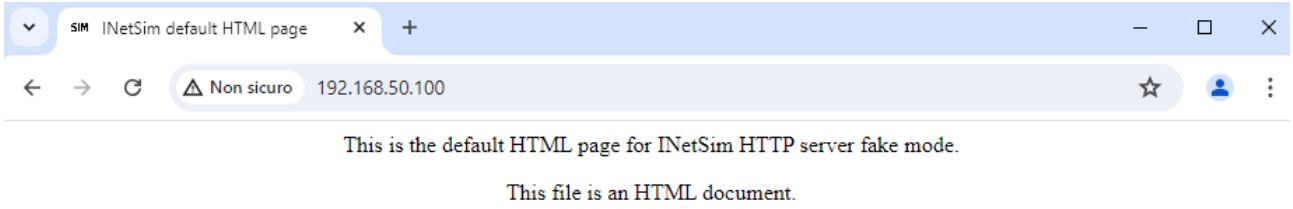
4) Navigazione HTTPS : apriamo il nostro browser internet su Windows, e inseriamo l indirizzo https://192.168.50.100, ottenendo una risposta dal server di Kali.



Attraverso l utilizzo di wireshark possiamo verificare che la comunicazione tra le 2 macchine è avvenuta nel modo corretto, il server di kali ha risposto dalla porta 443 utilizzando il protcollo TCP (syn,ack), per ricevere una risposta TLS (**T**ransport **L**ayer **S**ecurity).

19	0.053838093	192.168.50.102	192.168.50.100	TCP	66	49460 → 443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
20	0.053881012	192.168.50.100	192.168.50.102	TCP	66	443 → 49460 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM …
21	0.054374596	192.168.50.102	192.168.50.100	TCP	60	49460 → 443 [ACK] Seq=1 Ack=1 Win=65536 Len=0
22	0.054864420	192.168.50.102	192.168.50.100	TLSv1.3	1851	Client Hello
23	0.054877742	192.168.50.100	192.168.50.102	TCP	54	443 → 49460 [ACK] Seq=1 Ack=1798 Win=62464 Len=0
24	0.064628043	192.168.50.100	192.168.50.102	TCP	54	443 → 49459 [FIN, ACK] Seq=1444 Ack=2100 Win=64256 Len=0
25	0.065616814	192.168.50.102	192.168.50.100	TCP	60	49459 → 443 [ACK] Seq=2100 Ack=1445 Win=64256 Len=0
26	0.107955162	192.168.50.100	192.168.50.102	TLSv1.3	1497	Server Hello, Change Cipher Spec, Application Data, Application Data, …
27	0.109439978	192.168.50.102	192.168.50.100	TLSv1.3	134	Change Cipher Spec, Application Data
28	0.109440294	192.168.50.102	192.168.50.100	TLSv1.3	786	Application Data
29	0.115034468	192.168.50.100	192.168.50.102	TLSv1.3	309	Application Data
30	0.137511895	192.168.50.100	192.168.50.102	TLSv1.3	786	Application Data, Application Data, Application Data, Application Data
31	0.139804960	192.168.50.102	192.168.50.100	TCP	60	49460 → 443 [ACK] Seq=2610 Ack=2432 Win=65536 Len=0
32	0.140946678	192.168.50.102	192.168.50.100	TCP	60	49460 → 443 [FIN, ACK] Seq=2610 Ack=2432 Win=65536 Len=0
33	0.141304827	192.168.50.100	192.168.50.102	TCP	54	443 → 49460 [ACK] Seq=2432 Ack=2611 Win=61696 Len=0
34	0.166857923	192.168.50.102	192.168.50.100	TCP	66	49461 → 443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
35	0.166896997	192.168.50.100	192.168.50.102	TCP	66	443 → 49461 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM …
36	0.167854872	192.168.50.102	192.168.50.100	TCP	60	49461 → 443 [ACK] Seq=1 Ack=1 Win=65536 Len=0
37	0.169518604	192.168.50.102	192.168.50.100	TLSv1.3	2090	Client Hello
38	0.169595288	192.168.50.100	192.168.50.102	TCP	54	443 → 49461 [ACK] Seq=1 Ack=2037 Win=63104 Len=0
39	0.190033150	192.168.50.100	192.168.50.102	TLSv1.3	1497	Server Hello, Change Cipher Spec, Application Data, Application Data, …
40	0.191191759	192.168.50.102	192.168.50.100	TLSv1.3	84	Change Cipher Spec, Application Data
41	0.191192218	192.168.50.102	192.168.50.100	TCP	60	49461 → 443 [FIN, ACK] Seq=2067 Ack=1444 Win=64256 Len=0
42	0.192194108	192.168.50.102	192.168.50.100	TCP	66	49462 → 443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
43	0.192222171	192.168.50.100	192.168.50.102	TCP	66	443 → 49462 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM …
44	0.193167707	192.168.50.102	192.168.50.100	TCP	60	49462 → 443 [ACK] Seq=1 Ack=1 Win=65536 Len=0
45	0.193782743	192.168.50.102	192.168.50.100	TLSv1.3	2122	Client Hello
46	0.193800515	192.168.50.100	192.168.50.102	TCP	54	443 → 49462 [ACK] Seq=1 Ack=2069 Win=64000 Len=0
47	0.198301709	192.168.50.100	192.168.50.102	TCP	54	443 → 49461 [FIN, ACK] Seq=1444 Ack=2068 Win=64256 Len=0
48	0.199940530	192.168.50.102	192.168.50.100	TCP	60	49461 → 443 [ACK] Seq=2068 Ack=1445 Win=64256 Len=0
49	0.225872625	192.168.50.100	192.168.50.102	TLSv1.3	1497	Server Hello, Change Cipher Spec, Application Data, Application Data, …
50	0.227329096	192.168.50.102	192.168.50.100	TLSv1.3	134	Change Cipher Spec, Application Data
51	0.227329413	192.168.50.102	192.168.50.100	TLSv1.3	687	Application Data
52	0.227748960	192.168.50.100	192.168.50.102	TLSv1.3	309	Application Data
53	0.260666292	192.168.50.100	192.168.50.102	TLSv1.3	729	Application Data, Application Data, Application Data, Application Data
54	0.262015882	192.168.50.102	192.168.50.100	TCP	60	49462 → 443 [ACK] Seq=2782 Ack=2375 Win=65536 Len=0
55	0.263619566	192.168.50.102	192.168.50.100	TCP	60	49462 → 443 [FIN, ACK] Seq=2782 Ack=2375 Win=65536 Len=0

5) Navigazione HTTP : apriamo il nostro browser internet su Windows, e inseriamo l indirizzo http://192.168.50.100, ottenendo una risposta dal server di Kali.



Attraverso l utilizzo di wireshark possiamo verificare che la comunicazione tra le 2 macchine è avvenuta nel modo corretto, il server di kali ha risposto dalla porta 80 utilizzando il protcollo TCP (syn,ack),per ricevere una risposta HTTP.

1	0.000000000	192.168.50.102	192.168.50.100	TCP	66	49470 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
2	0.000027859	192.168.50.100	192.168.50.102	TCP	66	80 → 49470 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS…
3	0.000628941	192.168.50.102	192.168.50.100	TCP	60	49470 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0
4	0.001359693	192.168.50.102	192.168.50.100	TCP	66	49471 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
5	0.001377021	192.168.50.100	192.168.50.102	TCP	66	80 → 49471 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS…
6	0.001943331	192.168.50.102	192.168.50.100	HTTP	530	GET / HTTP/1.1

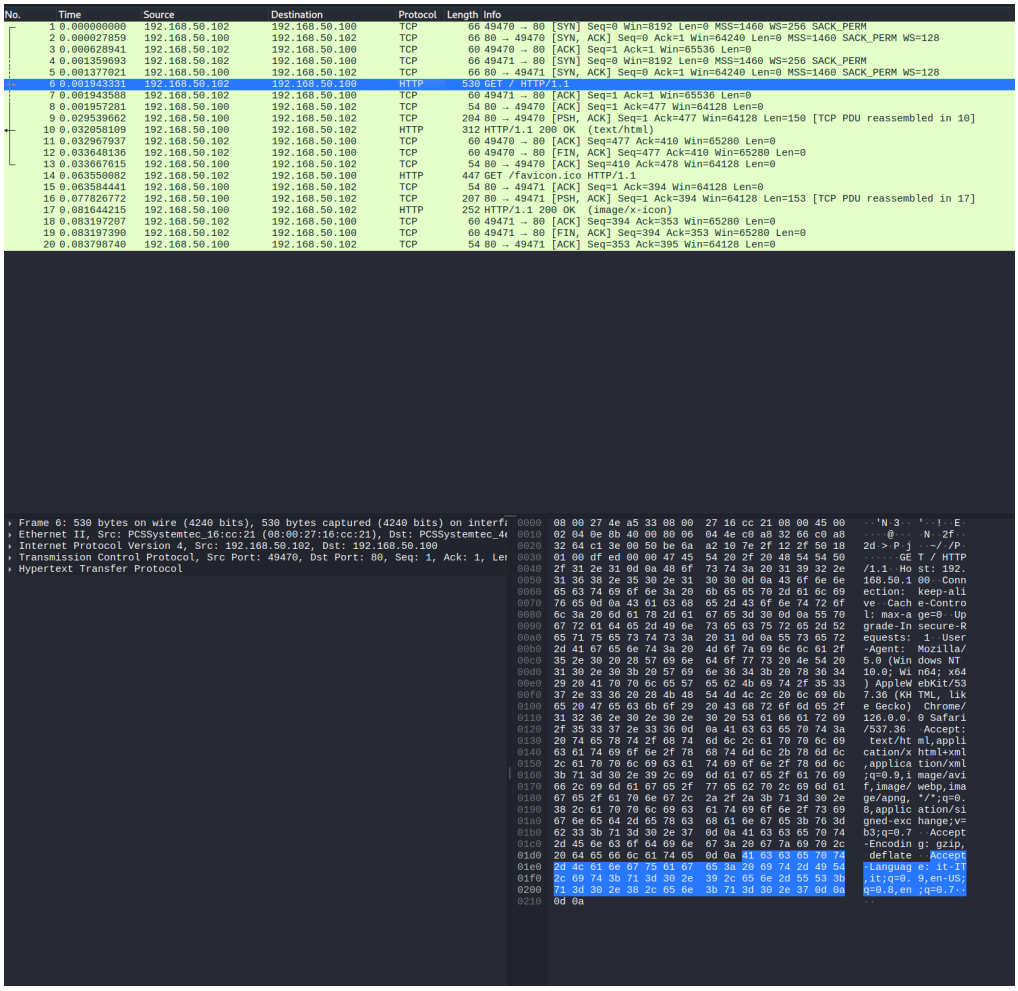
6) **Differenze** nella comunicazione **Http** e **Https**: come possiamo vedere dai dati raccolti su wireshark, le principali sono essenzialmente 2:

- **La porta di origine** : per Http è la 80, per Https è la 443

- **Utilizzo modello OSI in modo differente** : la richiesta HTTP sostanzialmente salta i livelli (Transport e Session), infatti essendo priva di criptatura e chiavi di accesso previsti nel metodo TLS che usa invece l HTTPS, risulta priva di informazioni riguardo alla sessione , e sopratutto leggibile da tutti.

Come possiamo vedere dalle 2 immagini qui sotto wireshark evidenzia in modo chiaro la differenza sopratutto in termini di cifratura, dove nella cattura dati Http, è possibile leggere tutte le informazioni relative alla pagina web e il suo contenuto, diversamente in quella Https, vedremo solo un sequenza di numeri e lettere apparentemente incomprensibili perché appunto criptati.

HTTP



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.50.102	192.168.50.100	TCP	66	49470 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
2	0.000027859	192.168.50.100	192.168.50.102	TCP	66	80 → 49470 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
3	0.000628941	192.168.50.102	192.168.50.100	TCP	60	49470 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0
4	0.001359693	192.168.50.102	192.168.50.100	TCP	66	49471 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
5	0.001377021	192.168.50.100	192.168.50.102	TCP	66	80 → 49471 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
6	0.001943331	192.168.50.102	192.168.50.100	HTTP	530	GET / HTTP/1.1
7	0.001943588	192.168.50.102	192.168.50.100	TCP	60	49471 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0
8	0.001957281	192.168.50.100	192.168.50.102	TCP	54	80 → 49470 [ACK] Seq=1 Ack=477 Win=64128 Len=0
9	0.029539662	192.168.50.100	192.168.50.102	TCP	204	80 → 49470 [PSH, ACK] Seq=1 Ack=477 Win=64128 Len=150 [TCP PDU reassembled in 10]
10	0.032058109	192.168.50.100	192.168.50.102	HTTP	312	HTTP/1.1 200 OK (text/html)
11	0.032967937	192.168.50.102	192.168.50.100	TCP	60	49470 → 80 [ACK] Seq=477 Ack=410 Win=65280 Len=0
12	0.033648136	192.168.50.102	192.168.50.100	TCP	60	49470 → 80 [FIN, ACK] Seq=477 Ack=410 Win=65280 Len=0
13	0.033667615	192.168.50.100	192.168.50.102	TCP	54	80 → 49470 [ACK] Seq=410 Ack=478 Win=64128 Len=0
14	0.063550082	192.168.50.102	192.168.50.100	HTTP	447	GET /favicon.ico HTTP/1.1
15	0.063584441	192.168.50.100	192.168.50.102	TCP	54	80 → 49471 [ACK] Seq=1 Ack=394 Win=64128 Len=0
16	0.077826772	192.168.50.100	192.168.50.102	TCP	207	80 → 49471 [PSH, ACK] Seq=1 Ack=394 Win=64128 Len=153 [TCP PDU reassembled in 17]
17	0.081644215	192.168.50.100	192.168.50.102	HTTP	252	HTTP/1.1 200 OK (image/x-icon)
18	0.083197207	192.168.50.102	192.168.50.100	TCP	60	49471 → 80 [ACK] Seq=394 Ack=353 Win=65280 Len=0
19	0.083197390	192.168.50.102	192.168.50.100	TCP	60	49471 → 80 [FIN, ACK] Seq=394 Ack=353 Win=65280 Len=0
20	0.083798740	192.168.50.100	192.168.50.102	TCP	54	80 → 49471 [ACK] Seq=353 Ack=395 Win=64128 Len=0

```
› Frame 6: 530 bytes on wire (4240 bits), 530 bytes captured (4240 bits) on interfa
› Ethernet II, Src: PCSSystemtec_16:cc:21 (08:00:27:16:cc:21), Dst: PCSSystemtec_4e
› Internet Protocol Version 4, Src: 192.168.50.102, Dst: 192.168.50.100
› Transmission Control Protocol, Src Port: 49470, Dst Port: 80, Seq: 1, Ack: 1, Len
› Hypertext Transfer Protocol

0000  08 00 27 4e a5 33 08 00  27 16 cc 21 08 00 45 00   ··'N·3·· '··!··E·
0010  02 04 0e 8b 40 00 80 06  04 4e c0 a8 32 66 c0 a8   ····@··· ·N··2f··
0020  32 64 c1 3e 00 50 be 6a  a2 10 7e 2f 12 2f 50 18   2d·>·P·j ··~/·/P·
0030  01 00 df ed 00 00 47 45  54 20 2f 20 48 54 54 50   ····· GE T / HTTP
0040  2f 31 2e 31 0d 0a 48 6f  73 74 3a 20 31 39 32 2e   /1.1··Ho st: 192.
0050  31 36 38 2e 35 30 2e 31  30 30 0d 0a 43 6f 6e 6e   168.50.1 00··Conn
0060  65 63 74 69 6f 6e 3a 20  6b 65 65 70 2d 61 6c 69   ection:  keep-ali
0070  76 65 0d 0a 43 61 63 68  65 2d 43 6f 6e 74 72 6f   ve··Cach e-Contro
0080  6c 3a 20 6d 61 78 2d 61  67 65 3d 30 0d 0a 55 70   l: max-a ge=0··Up
0090  67 72 61 64 65 2d 49 6e  73 65 63 75 72 65 2d 52   grade-In secure-R
00a0  65 71 75 65 73 74 73 3a  20 31 0d 0a 55 73 65 72   equests:  1··User
00b0  2d 41 67 65 6e 74 3a 20  4d 6f 7a 69 6c 6c 61 2f   -Agent:  Mozilla/
00c0  35 2e 30 20 28 57 69 6e  64 6f 77 73 20 4e 54 20   5.0 (Win dows NT
00d0  31 30 2e 30 3b 20 57 69  6e 36 34 3b 20 78 36 34   10.0; Wi n64; x64
00e0  29 20 41 70 70 6c 65 57  65 62 4b 69 74 2f 35 33   ) AppleW ebKit/53
00f0  37 2e 33 36 20 28 4b 48  54 4d 4c 2c 20 6c 69 6b   7.36 (KH TML, lik
0100  65 20 47 65 63 6b 6f 29  20 43 68 72 6f 6d 65 2f   e Gecko)  Chrome/
0110  31 32 36 2e 30 2e 30 2e  30 20 53 61 66 61 72 69   126.0.0. 0 Safari
0120  2f 35 33 37 2e 33 36 0d  0a 41 63 63 65 70 74 3a   /537.36··Accept:
0130  20 74 65 78 74 2f 68 74  6d 6c 2c 61 70 70 6c 69    text/ht ml,appli
0140  63 61 74 69 6f 6e 2f 78  68 74 6d 6c 2b 78 6d 6c   cation/x html+xml
0150  2c 61 70 70 6c 69 63 61  74 69 6f 6e 2f 78 6d 6c   ,applica tion/xml
0160  3b 71 3d 30 2e 39 2c 69  6d 61 67 65 2f 61 76 69   ;q=0.9,i mage/avi
0170  66 2c 69 6d 61 67 65 2f  77 65 62 70 2c 69 6d 61   f,image/ webp,ima
0180  67 65 2f 61 70 6e 67 2c  2a 2f 2a 3b 71 3d 30 2e   ge/apng, */*;q=0.
0190  38 2c 61 70 70 6c 69 63  61 74 69 6f 6e 2f 73 69   8,applic ation/si
01a0  67 6e 65 64 2d 65 78 63  68 61 6e 67 65 3b 76 3d   gned-exc hange;v=
01b0  62 33 3b 71 3d 30 2e 37  0d 0a 41 63 63 65 70 74   b3;q=0.7··Accept
01c0  2d 45 6e 63 6f 64 69 6e  67 3a 20 67 7a 69 70 2c   -Encodin g: gzip,
01d0  20 64 65 66 6c 61 74 65  0d 0a 41 63 63 65 70 74    deflate··Accept
01e0  2d 4c 61 6e 67 75 61 67  65 3a 20 69 74 2d 49 54   -Languag e: it-IT
01f0  2c 69 74 3b 71 3d 30 2e  39 2c 65 6e 2d 55 53 3b   ,it;q=0. 9,en-US;
0200  71 3d 30 2e 38 2c 65 6e  3b 71 3d 30 2e 37 0d 0a   q=0.8,en ;q=0.7··
0210  0d 0a                                              ··
```

HTTPS

15	0.035413168	192.168.50.102	192.168.50.100	TCP	60	49472 → 443 [ACK] Seq=2036 Ack=1445 Win=64256 Len=0
16	0.036674256	192.168.50.100	192.168.50.102	TLSv1.3	1497	Server Hello, Change Cipher Spec, Application Data, Application Data, Application Data…
17	0.037553131	192.168.50.102	192.168.50.100	TLSv1.3	84	Change Cipher Spec, Application Data
18	0.037553664	192.168.50.102	192.168.50.100	TCP	60	49473 → 443 [FIN, ACK] Seq=1732 Ack=1444 Win=64256 Len=0
19	0.038234152	192.168.50.102	192.168.50.100	TCP	66	49474 → 443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
20	0.038256447	192.168.50.100	192.168.50.102	TCP	66	443 → 49474 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
21	0.039345122	192.168.50.102	192.168.50.100	TCP	60	49474 → 443 [ACK] Seq=1 Ack=1 Win=65536 Len=0
22	0.041841779	192.168.50.102	192.168.50.100	TLSv1.3	1755	Client Hello
23	0.041870135	192.168.50.100	192.168.50.102	TCP	54	443 → 49474 [ACK] Seq=1 Ack=1702 Win=62592 Len=0
24	0.049082029	192.168.50.100	192.168.50.102	TCP	54	443 → 49473 [FIN, ACK] Seq=1444 Ack=1733 Win=62592 Len=0
25	0.049688965	192.168.50.102	192.168.50.100	TCP	60	49473 → 443 [ACK] Seq=1733 Ack=1445 Win=64256 Len=0
26	0.067932426	192.168.50.100	192.168.50.102	TLSv1.3	1497	Server Hello, Change Cipher Spec, Application Data, Application Data, Application Data…
27	0.069496534	192.168.50.102	192.168.50.100	TLSv1.3	134	Change Cipher Spec, Application Data
28	0.069497147	192.168.50.102	192.168.50.100	TLSv1.3	786	Application Data
29	0.070349056	192.168.50.100	192.168.50.102	TLSv1.3	309	Application Data
30	0.096787796	192.168.50.100	192.168.50.102	TLSv1.3	786	Application Data, Application Data, Application Data, Application Data
31	0.098414006	192.168.50.102	192.168.50.100	TCP	60	49474 → 443 [ACK] Seq=2514 Ack=2432 Win=65536 Len=0
32	0.099918985	192.168.50.102	192.168.50.100	TCP	60	49474 → 443 [FIN, ACK] Seq=2514 Ack=2432 Win=65536 Len=0
33	0.099940714	192.168.50.100	192.168.50.102	TCP	54	443 → 49474 [ACK] Seq=2432 Ack=2515 Win=61824 Len=0
34	0.119823478	192.168.50.102	192.168.50.100	TCP	66	49475 → 443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
35	0.119854106	192.168.50.100	192.168.50.102	TCP	66	443 → 49475 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
36	0.120366189	192.168.50.102	192.168.50.100	TCP	60	49475 → 443 [ACK] Seq=1 Ack=1 Win=65536 Len=0
37	0.120717726	192.168.50.102	192.168.50.100	TLSv1.3	2058	Client Hello
38	0.120733706	192.168.50.100	192.168.50.102	TCP	54	443 → 49475 [ACK] Seq=1 Ack=2005 Win=62336 Len=0
39	0.141638147	192.168.50.100	192.168.50.102	TLSv1.3	1497	Server Hello, Change Cipher Spec, Application Data, Application Data, Application Data…
40	0.143646299	192.168.50.102	192.168.50.100	TLSv1.3	84	Change Cipher Spec, Application Data
41	0.143646676	192.168.50.102	192.168.50.100	TCP	60	49475 → 443 [FIN, ACK] Seq=2035 Ack=1444 Win=64256 Len=0
42	0.143646727	192.168.50.102	192.168.50.100	TCP	66	49476 → 443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
43	0.143708245	192.168.50.100	192.168.50.102	TCP	66	443 → 49476 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
44	0.144569568	192.168.50.102	192.168.50.100	TCP	60	49476 → 443 [ACK] Seq=1 Ack=1 Win=65536 Len=0
45	0.145243974	192.168.50.102	192.168.50.100	TLSv1.3	2122	Client Hello
46	0.145255984	192.168.50.100	192.168.50.102	TCP	54	443 → 49476 [ACK] Seq=1 Ack=2069 Win=64000 Len=0
47	0.150630740	192.168.50.100	192.168.50.102	TCP	54	443 → 49475 [FIN, ACK] Seq=1444 Ack=2036 Win=64000 Len=0
48	0.151421026	192.168.50.102	192.168.50.100	TCP	60	49475 → 443 [ACK] Seq=2036 Ack=1445 Win=64256 Len=0
49	0.167517100	192.168.50.100	192.168.50.102	TLSv1.3	1497	Server Hello, Change Cipher Spec, Application Data, Application Data, Application Data…
50	0.171640913	192.168.50.102	192.168.50.100	TLSv1.3	134	Change Cipher Spec, Application Data
51	0.171641329	192.168.50.102	192.168.50.100	TLSv1.3	687	Application Data
52	0.172149752	192.168.50.100	192.168.50.102	TLSv1.3	309	Application Data
53	0.196232599	192.168.50.100	192.168.50.102	TLSv1.3	729	Application Data, Application Data, Application Data, Application Data
54	0.196729678	192.168.50.102	192.168.50.100	TCP	60	49476 → 443 [ACK] Seq=2782 Ack=2375 Win=65536 Len=0
55	0.198108549	192.168.50.102	192.168.50.100	TCP	60	49476 → 443 [FIN, ACK] Seq=2782 Ack=2375 Win=65536 Len=0
56	0.198130499	192.168.50.100	192.168.50.102	TCP	54	443 → 49476 [ACK] Seq=2375 Ack=2783 Win=64512 Len=0

```
▸ Frame 22: 1755 bytes on wire (14040 bits), 1755 bytes captured (14040 bits) on ir
▸ Ethernet II, Src: PCSSystemtec_16:cc:21 (08:00:27:16:cc:21), Dst: PCSSystemtec_4e
▸ Internet Protocol Version 4, Src: 192.168.50.102, Dst: 192.168.50.100
▸ Transmission Control Protocol, Src Port: 49474, Dst Port: 443, Seq: 1, Ack: 1, Le
▸ Transport Layer Security
```

```
0000  08 00 27 4e a5 33 08 00  27 16 cc 21 08 00 45 00   ··'N·3·· '··!··E·
0010  06 cd 10 97 40 00 80 06  fd 78 c0 a8 32 66 c0 a8   ····@··· ·x··2f··
0020  32 64 c1 42 01 bb 5c 49  f8 f9 91 e7 be 9d 50 18   2d·B··\I ######··P·
0030  01 00 ec da 00 00 16 03  01 06 a0 01 00 06 9c 03   ######## ########
0040  03 26 fa 9b 3c 80 72 8c  6f ff e1 e0 a3 5f e0 ca   ·&··<·r· o····_··
0050  78 05 34 43 76 99 4f d4  79 28 bd 44 da d6 bb ff   x·4Cv·O· y(·D····
0060  06 20 b0 56 a1 df 4c 84  fd ed 73 36 cc 61 10 6a   ··· V··L· ··s6·a·j
0070  c5 1d 70 c8 7b ad d5 18  c2 f3 46 5a 3d 8c 4e 5b   ··p·{··· ··FZ=·N[
0080  e1 c4 00 20 9a 9a 13 01  13 02 13 03 c0 2b c0 2f   ######## #####+·/
0090  c0 2c c0 30 cc a9 cc a8  c0 13 c0 14 00 9c 00 9d   ·,·0···· ########
00a0  00 2f 00 35 01 00 06 33  da da 00 00 00 10 00 0e   ·/·5··3· ########
00b0  00 0c 02 68 32 08 68 74  74 70 2f 31 2e 31 00 05   ···h2·ht tp/1.1··
00c0  00 05 01 00 00 00 00 00  0b 00 02 01 00 00 0a 00   ######## ########
00d0  0c 00 0a fa fa 63 99 00  1d 00 17 00 18 00 2b 00   #####c·· ######+·
00e0  07 06 aa aa 03 04 03 03  00 23 00 00 fe 0d 00 ba   ######## ·#######
00f0  00 00 01 00 01 b3 00 20  05 59 24 c5 b6 fc 61 aa   ######## ·Y$···a·
0100  6e e5 6b 44 ce 67 3d ea  dd 88 76 a1 5b bc e3 fc   n·kD·g=· ··v·[···
0110  b6 8a 5b 7f 4c e4 9f 48  00 90 d4 1e b7 d5 2f dd   ··[·L··H ######/·
0120  f1 16 87 61 3f 82 14 eb  02 00 a0 d7 11 b5 4e 2f   ···a?··· ######N/
0130  61 33 b9 e7 66 01 5d fe  61 8a fc 3b ee 38 dc c9   a3··f·]· a··;·8··
0140  37 7a 0b 25 10 eb 96 0d  43 a1 6d d1 fc d5 62 b6   7z·%···· C·m···b·
0150  f8 23 13 08 1e 42 12 83  79 58 69 d3 86 f3 43 24   ·#···B·· yXi···C$
0160  0b fe 8a 65 16 38 e3 1d  a0 3c 75 fe 1c 2b 5b 53   ···e·8·· ·<u··+[S
0170  4a 64 9d c1 60 ce 43 1a  20 70 30 82 16 9b 5c 39   Jd··`·C· ·p0···\9
0180  02 c6 19 96 e9 5f 58 f5  27 a3 ab 2a 81 9c 8d aa   ######_X ·'··*····
0190  05 9d 14 7c 87 60 d1 6f  56 c8 00 57 da 41 90 f5   ···|·`·o V··W·A··
01a0  3c f7 ee ef d6 4e c6 58  68 78 00 17 00 00 ff 01   <····N·X hx######
01b0  00 01 00 44 69 00 05 00  03 02 68 32 00 0d 00 12   ···Di··· ··h2····
01c0  00 10 04 03 08 04 04 01  05 03 08 05 05 01 08 06   ######## ########
01d0  06 01 00 1b 00 03 02 00  02 00 2d 00 02 01 01 00   ######## ##-#####
01e0  33 04 ef 04 ed fa fa 00  01 00 63 99 04 c0 b2 00   3####### ##c#####
01f0  51 da 47 26 d6 31 6e 0c  2c cb 5b a8 69 12 e4 91   Q·G&·1n· ,·[·i···
0200  c1 75 c3 4e e6 78 87 94  f7 81 4f 6c 04 03 dd fa   ·u·N·x·· ··Ol····
0210  11 02 8c b9 4c 17 7d eb  28 5c 75 31 92 0a 47 2f   ····L·}· (\u1··G/
0220  d2 d4 bb 39 60 03 99 aa  ce 36 8a 2c 5f 24 5c 19   ···9`··· ·6·,_$\·
0230  fa aa 91 1b a7 4f 92 0b  c9 cc 6b 8d e8 44 85 51   ######O· ··k··D·Q
0240  5d 76 f1 4f 1d 20 24 21  17 b3 74 4a 84 b7 25 be   ]v·O·$! ··tJ··%·
0250  5d b2 a3 3d 64 54 9b 5b  5f af 7b c4 23 bc 12 c1   ]··=dT·[ _·{·#···
0260  01 68 c3 25 7a c1 07 3e  3f eb a2 c6 52 30 c2 ba   ·h·%z··> ?···R0··
0270  1f 99 31 00 76 a4 4a f2  c7 7e 0c f3 9a ea 20 3f   ····1·v·J· ·~···· ?
0280  b7 4b 65 35 f2 40 75 11  4e a2 66 75 77 d7 cb f8   ·Ke5·@u· N·fuw···
0290  f8 98 5a 58 98 df 49 84  6f 92 4b 1b a8 7f 3d 13   ····ZX·· I··o·K·· ··=·
02a0  33 78 c2 72 9e 23 3b 1e  84 b2 62 7a c2 c2 56 67   3x·r·#;· ··bz··Vg
02b0  de 21 3a af 2b 22 14 4a  60 3d c5 51 8b 5a 1d c5   ·!:·+"·J `=·Q·Z··
```